SUEZ AI Gateway MVP – Implementation Plan

Phase 0 — Project Foundations & Local Runtime

- Repository setup with mono-repo or multi-repo structure.
- Create services: gateway, control-plane, portal-ui, postgres, redis, opa, policy-llm.
- Docker Compose configuration and service networking.
- Database schema, migrations, and seed data.
- Gateway, Control Plane, and Portal UI skeleton services.
- OPA container and initial Rego policy bundle.
- Local LLM container for policy generation.
- Exit: docker compose up runs all services and UI loads.

Phase 1 — Authentication & Identity Context

- Implement API key authentication middleware.
- Optional JWT authentication.
- Normalize requests to IdentityContext structure.
- Control-plane endpoints for tenants and API keys.
- UI screens for login, tenant management, and API key generation.
- Exit: authenticated requests produce IdentityContext.

Phase 2 — Access Workflow

- Access request APIs: create, list, approve, deny.
- Grant model and creation after approval.
- Audit logging for requests and approvals.
- UI screens for request submission, approval queue, and grants view.
- Exit: user requests access and admin approves via portal.

Phase 3 — Gateway Core Enforcement

- Implement POST /v1/chat/completions endpoint.
- Provider abstraction with OpenAI-compatible adapter.
- OPA policy evaluation integration.
- Redis quota enforcement (RPM/TPM).

- Token estimation before and reconciliation after provider call.
- Audit logs for model requests.
- Exit: authorized requests succeed, quota and policy denials enforced.

Phase 4 — Tool Governance

- Tool registry APIs for creation and management.
- Tool execution endpoint POST /v1/tools/execute.
- JSON schema validation for tool input.
- OPA authorization checks before execution.
- Audit logs for tool calls.
- UI screens for tool registry and tool access requests.

Phase 5 — Natural Language Policy Builder

- Policy draft model for storing natural language rules and generated Rego.
- Policy generation service calling local LLM.
- Policy generation API endpoint.
- Rego syntax validation.
- UI policy builder screen and draft review page.

Phase 6 — Policy Testing & Lifecycle

- Policy test runner for validating allow/deny cases.
- Publish and rollback endpoints.
- OPA bundle updates for published policies.
- UI screens for test results and policy versions.
- Exit: tests must pass before policy publish.

Phase 7 — Audit & Observability UI

- Audit APIs for requests, tools, and policies.
- Filtering by actor, model, tool, decision, trace ID.
- UI audit log pages and trace detail views.
- Enhanced structured logging.

Phase 8 — Local Operations Hardening

- TLS for local environment.
- Secure secrets and DB permissions.
- Backup and export scripts.
- Seed demo data for tenants, models, tools, and policies.
- Documentation for setup and architecture.